

Rajalakshmi Engineering College

Department of Artificial Intelligence & Machine Learning

III Year (2025 – 2026) - AI23521: Build and Deployment of ML app

Mini Project - Abstract

Title	Sentiment Analysis of Amazon Fine Food Reviews	
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Project ID		

This project presents a complete end-to-end pipeline for analyzing customer sentiments from the Amazon Fine Food Reviews dataset, encompassing data preprocessing, feature engineering, supervised machine learning, model evaluation, and deployment via a RESTful API. The primary objective is to develop a predictive system capable of accurately classifying textual product reviews into positive or negative sentiments, providing actionable insights for businesses, customers, and researchers interested in consumer feedback analysis.

The project implements multiple supervised learning algorithms to perform sentiment classification, including Logistic Regression, Support Vector Machines (SVM), Random Forests, and Gradient Boosting. Each model is trained on a carefully split training dataset and evaluated using standard performance metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices to ensure robustness and generalization to unseen data. Comparative analysis of the models allows selection of the most effective algorithm for deployment.

For real-world usability, the trained model is deployed as a RESTful API using Flask, enabling external clients to send textual reviews and receive sentiment predictions in real-time. The deployment is further enhanced using Docker containerization, which encapsulates the application, dependencies, and runtime environment into a portable and reproducible package. This ensures that the application can run seamlessly across different systems without compatibility issues. Additionally, the project explores optional frontend integration using simple HTML/JavaScript interfaces for interactive prediction, demonstrating end-to-end usability.

This project serves as a practical demonstration of combining natural language processing, supervised machine learning, and modern deployment techniques to deliver a scalable and reliable application. It highlights the importance of proper data preprocessing, feature engineering, model evaluation, and containerized deployment for building production-ready AI solutions, offering significant value for businesses seeking to analyze large-scale consumer reviews efficiently.

SUPERVISOR

REVIEWER